Gamified Learning for Rural Education

Design Thinking for innovation & Entrepreneurship (DTIE)

Section : 1

Project Batch-9A

Team Members – Group 9A

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Vision and Mission

Our Vision :

Bridge the learning poverty gap across 600,000+ Indian villages by making foundational literacy and numeracy as engaging as play.

Our Mission :

Deploy zero-data, multi-lingual gamified micro-modules on low-cost smartphones to boost retention and mastery by 3x by 2028.

Problem Statement

53% : Grade 5 rural students unable to read standard Grade 2 texts due to passive rote learning methods.

46.1% : Rural primary schools lacking stable internet connectivity, breaking online EdTech models.

35% : Unengaging traditional curriculum causes low motivation and high early school dropouts.

Research / Facts / Market & Industry Study

Market Overview: India Gamified EdTech market expanding from $50M in 2025 to $204M by 2031 (CAGR of 26.4%).

Key Points:

- **Addressable Rural Learners:** 295 Million primary/secondary K-12 students.
- **Mobile-First Delivery Share:** 73% market adoption in rural households.
- **National Mobile Connectivity Base:** 969+ Million total internet subscriptions expanding digital reach.

Target User - Customer

Primary Learner : Age 6-12 · Rural Primary

Motivated by game quests, rewards, regional avatars, and instant visual feedback.

Rural Educator : Government School Teacher

Needs automated progress tracking and offline lesson assistance without technical complexity.

Competitors / Benchmarks

Some of the common benchmarks are:

• VidyaSetu: The leading benchmark for gamified user onboarding. Its landing page immediately pulls the user into a 5-second trial game before requiring registration, an approach highly effective for low-literacy or first-time digital users.

• Gimkit: Exceptional benchmarks for dashboard simplification. They showcase how game pin-codes and minimal text entry can instantly launch multiplayer or single-player educational modes.

Risks and Gap

- **Hardware Constraints:** Outdated Android hardware with limited internal memory causing app crashes. *Mitigation:* Ultra-lightweight vector graphics engine maintaining build size under 15MB.

- **Teacher Resistance:** Fear of digital replacement or lack of digital literacy among staff. *Mitigation:* Teacher co-pilot interface framing educators as quest masters.

- **Power Outages:** Frequent power cuts interrupting daily school activity. *Mitigation:* Instant auto-save state after every individual micro-quest module.

Concept/Idea/Prototype/Product/Solution

Our idea is to create a website that:

- **1. Works Without Internet :** Students can share scores, badges, and progress with nearby classmates using simple device-to-device connections without needing internet.

- **2. Fun Math & Reading Games :** Simple mini-games that automatically adjust difficulty—getting easier or harder based on how well the child answers.

- **3. Local Stories & Characters :** Lessons designed using familiar village folk tales and regional stories so children connect with the content instantly.

Images/Storyboard/Architecture



1.Local Wi-Fi Mesh: Zero-cost hotspot connecting student devices within classroom range.
2.Student Tablet App: Interactive game modules, badges, and adaptive practice.
3.Sync Dashboard: Weekly aggregate analytics sync when the teacher connects to a mobile network zone.

Lean Canvas Model

- **Problem:** Low foundational literacy, high absenteeism, lack of rural connectivity.
- **Solution:** Offline gamified learning application running on shared local school hardware.
- **Unique Value Proposition:** Zero-data vernacular game loops boosting FLN retention by 3x.
- **Unfair Advantage:** Proprietary lightweight mesh-sync protocol tailored for offline rural areas.
- **Customer Segments:** State Education Departments, Rural NGOs, Corporate CSR Funds, Village Primary Schools.
- **Cost Structure:** Vernacular Content Localization, Mobile App Engineering, School Hub Box Hardware.
- **Revenue Streams:** Annual Per-School Licensing Model, NGO Deployment Grants, Government B2G Contracts.
- **Channel:** Through offline platform with a gamified learning experience.

Capital (Financial)

Fund Deployment Breakdown:

- App Engineering & Offline Mesh Protocol: **40%**
- Regional Content & 12 Vernacular Languages: **30%**
- Pilot School Field Deployment: **20%**
- Operations & Administration: **10%**

Unit Economics per School:

- Hardware Setup Cost: ₹4,500 / school
- Annual SaaS Licensing Contract: ₹12,000 / year
- Payback Period: 4.5 Months
- Year 3 Projected Revenue: ₹1.8 Crore

Roadmap/Timeline

A realistic 12-month implementation roadmap for a rural-focused gamified learning web platform balances technical execution with community-driven validation. In low-resource settings, standard tech deployment timelines must lengthen to accommodate field trials, hardware compatibility syncs, and offline caching tests.

Phase 1: Research and Foundation
Phase 2: Prototype and Caching
Phase 3: Pilot trails and Translations
Phase 4: Production Launch